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Key Facts

Nation	United States of America (USA)
Objective(s)	Solar Orbit; Planet Hunter
Spacecraft	Kepler
Spacecraft Mass	2,291 pounds (1,039 kilograms)
Mission Design and Management	NASA / ARC / JPL
Launch Vehicle	Delta 7925-10L (no. D339)
Launch Date and Time	March 6, 2009 / 03:49:57 UTC March 7
Launch Site	Cape Canaveral, Fla. / SLC-17B
Scientific Instruments	Photometer (Schmidt Telescope)

Key Dates

March 6, 2009: Launch

May 13, 2009: Kepler begins its operational mission

December 2011: NASA announces Kepler has found the first planet, Kepler-22b, in the habitable zone of a star outside our solar system

Oct. 30, 2018: NASA announces Kepler is out of fuel and will be retired in its current orbit

In Depth

NASA's Kepler, the 10th in a series of low-cost, low-development-time and highly focused Discovery-class science missions, was designed to discover Earth-like planets orbiting other stars in our region of the Milky Way.

The spacecraft was named after the famed German astronomer Johannes Kepler (1571-1630).

Kepler was equipped to look for planets with size spans from one-half to twice the size of Earth (terrestrial planets) in the habitable zone of their stars where liquid water might exist in the natural state on the surface of the planet.

Its scientific goals included determining the abundance of these planets and the distribution of sizes and shapes of their orbits, estimating the number of planets in multiple-star systems, and determining the properties of stars that have planetary systems.

Kepler detected planets by observing transits, or tiny dips in the brightness of a star that occur when a planet crosses in front of the star.

The spacecraft was basically a single instrument—a specially designed 3-foot (1-meter) diameter aperture telescope and image sensor array—with a spacecraft built around it. The diameter of the telescope's mirror was 4 feet, 7 inches (1.4 meters), one of the largest mirrors beyond Earth orbit.

Kepler was designed to monitor about 100,000 main-sequence stars over a period of three-and-a-half years.

Kepler was launched at 10:49 p.m. March 6, 2009 EST (03:49:57 UTC March 7, 2009), into an initial Earth orbit at 115×115 miles (185×185 kilometers) at a 28.5-degree inclination. After another first stage burn, the second stage fired to set Kepler on an escape trajectory into solar orbit. It passed lunar orbit at 04:20 UT March 9, eventually entering heliocentric orbit at 0.97×1.041 AU at 0.5-degree inclination to the solar ecliptic.

In order to improve resolution, on April 23, mission planners optimized the focus of the telescope by moving the primary mirror 40 micrometers toward the focal plane and tilting it by 0.0072 degrees.

Less than a month later, on May 13, Kepler finished its commissioning and began its operational mission. During its first six weeks of operation, Kepler discovered five exoplanets—named Kepler 4b, 5b, 6b, 7b and 8b (which NASA announced in January 2010).

In April 2010, mission scientists published results that showed that Kepler had discovered the first confirmed planetary system with more than one planet transiting the same star, Kepler-9. That discovery was the result of surveying more than 156,000 stars over a period of seven months.

The planetary system orbiting Kepler-11, a yellow dwarf star about 2,000 light years from Earth, included six planets. NASA announced in February 2011 that these planets were larger than Earth, with the largest ones comparable in size to Uranus and Neptune.

In 2011, Kepler suffered at least two safe mode events. The spacecraft essentially shut down science operations as a result of a suspected anomaly. In both cases, in February and March, the Kepler project team was able to revive the vehicle relatively quickly, within two to three days.

In September, mission scientists announced the discovery of a planet, Kepler-16b, orbiting two stars, where we might expect a double sunset, much like the fictional planet Tatooine depicted in the film "Star Wars." A subsequent double-star system was announced in January 2012 and multiple planets orbiting multiple stars—the Kepler-47 system—was announced in August 2012.

In December 2011, NASA announced that Kepler had found its first planet, Kepler-22b, in the habitable zone of a star.

In April 2012, the mission, closing in on its three-and-a-half-year lifetime, was formally extended through fiscal year 2016 after a review of its operations. The extended mission started Nov. 15, 2012. By that time, Kepler had identified more than 2,300 planet candidates and confirmed more than 100 planets.

Based on data collected by Kepler, scientists were able to announce in January 2013 that about 17% of stars (about one-sixth) have an Earth-sized planet in an orbit closer than Mercury is to our Sun. Given that the Milky Way has about 100 billion stars, this would suggest at least 17 billion Earth-sized worlds in our galaxy. (In November 2013, this number was revised up to 40 billion).

Following two brief lapses into safe mode in May 2013, one of the spacecraft's four reaction wheels (No. 4) was found to have failed. Given that an earlier one failed in July 2012 and that at least three such wheels were needed to accurately aim the telescope, there was anxiety that the mission might be jeopardized.

After this and another safe mode event in late May, Kepler operated in Point Rest State (PRS) mode—where the spacecraft used thrusters and solar pressure to control pointing—while controllers devised a way to reactivate the wheels necessary for accurate pointing of the spacecraft.

On Aug. 15, 2013, NASA officially announced it was ending efforts to fully recover Kepler. NASA solicited proposals from the public on how to reformulate a new mission for Kepler given its obvious limitations.

During this period, in October 2013, Kepler mission scientists announced that they had conclusively identified the first Earth-sized rocky planet, Kepler-78b, which circles its host star every eight-and-a-half years, making it a very hot planet. (A later announcement in April 2014 confirmed the discovery of the first Earth-sized planet, Kepler-186f, in the habitable zone of a star.)

At the end of 2013, the Kepler team proposed a new mission, known as K2 (Second Light), using the two remaining reaction wheels to investigate smaller and dimmer red dwarf stars.

Mission definition of the K2 proposal continued into 2014, with the mission being approved by NASA in May 2014 and data collection beginning May 30. Observations continued through the year with several campaigns of data collection.

By January 2015, Kepler had found 1,004 confirmed exoplanets in about 400 star systems.

By November 2016, Kepler was in its 11th campaign of scientific observation, which began Sept. 24, 2016.

On May 22, 2017, using data from Kepler's extended K2 mission, astronomers pinned down the orbital period of the outermost planet in the famous TRAPPIST-1 system—home to seven Earth-size planets. The data backed up the theory that the planets likely migrated inward during the system's formation.

On Dec. 14, 2017, an eighth planet was found in the Kepler-90 system, equal to our own solar system in having the largest number of known planets. All of the planets crowd closer to their star than Earth to our Sun. The discovery was made, in part, using artificial intelligence.

In January 2018, an Australian car mechanic sifting through K2's data discovered a four-planet system with Neptune-size worlds. The discovery was a highlight among the contributions of citizen scientists who combed through immense amounts of K2 data in search of exoplanets. Scientists went on to find a fifth planet in the system.

In May 2018, Kepler finished six months of supernovae observations. The spacecraft captured the beginning stages of the stellar explosions with unprecedented precision to help resolve a longstanding mystery: What sets them off?

On Oct. 30, 2018, after nine years in deep space collecting data, NASA announced that Kepler had run out of fuel. The spacecraft was retired in its current, safe orbit, away from Earth.

Kepler left behind a legacy of more than 2,600 planet discoveries from outside our solar system, many of which could be promising places for life.

"When we started conceiving this mission 35 years ago we didn't know of a single planet outside our solar system," said the Kepler mission's founding principal investigator, William Borucki. "Now that we know planets are everywhere, Kepler has set us on a new course that's full of promise for future generations to explore our galaxy."

Before retiring the spacecraft, scientists pushed Kepler to its full potential, successfully completing multiple observation campaigns and downloading valuable science data even after initial warnings of low fuel. The data will complement the data from TESS, the Transiting Exoplanet Survey Satellite, launched April 18, 2018.

Key Source

Siddiqi, Asif A. [Beyond Earth: A Chronicle of Deep Space Exploration, 1958-2016](#). NASA History Program Office, 2018.

